6TH GRADE MATH · UNIT 2: OPERATIONS OF FRACTIONS AND DECIMALS

Quilt Shop Makers

Measure, cut, and price fabric to sew a class quilt.

CP2

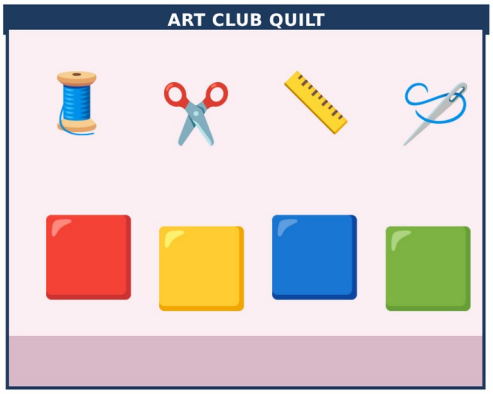
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The art club is sewing a quilt to sell at the school auction. Fabric is sold by the yard in fractions of a yard, squares must be cut to exact sizes, and every material has a price in dollars and cents.

You will use fraction and decimal operations to plan the cutting, figure out how much fabric to buy, and find the total cost of your quilt.



Fabric Rules

Sold by the ¼ yard

1 yard = 36 inches

Quilt Size

4 squares across

5 squares down

Learning Goals

- I can add and subtract fractions and mixed numbers to measure fabric.
- I can multiply and divide fractions to cut pieces.
- I can multiply and add decimals to find costs.

Project Timeline

When	What you will do
Day 1	Measure the Fabric
Day 2	Cut the Squares
Day 3	Price the Quilt
Final	Quilt Plan & Price Sheet + reflection

Part 1: Measure the Fabric

Day 1 · Each quilt square is cut from strips of fabric.

Fabric Price Cards

Red Cotton	Yellow Print	Blue Denim
$7.60 per yard	$8.40 per yard	$9.25 per yard
44 in wide	44 in wide	58 in wide

Green Floral	Batting	Thread
$6.80 per yard	$4.50 per yard	$2.35 per spool
44 in wide	90 in wide	250 yards

1. A strip is $\frac{3}{4}$ yard long. You cut off $\frac{1}{4}$ yard. How much is left?

   Starter: $\frac{3}{4} - \frac{1}{4} =$ ___

2. You have $1\frac{1}{2}$ yards of red and $\frac{3}{4}$ yard of yellow. How many yards is that in all?

3. Blue fabric comes in a $2\frac{1}{3}$ yard piece. You use $1\frac{2}{3}$ yards. How much is left?

4. Which is longer, $\frac{5}{8}$ yard or $\frac{2}{3}$ yard? Explain how you know.

Part 2: Cut the Squares

Day 2 · Each square needs $\frac{1}{8}$ yard of fabric.

5. How many squares can you cut from $\frac{3}{4}$ yard of fabric?

Starter: $\frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times ___ = ___$

6. The quilt has 20 squares. How many yards of fabric do you need?

7. You want $\frac{1}{2}$ of the squares to be red. How many yards of red fabric is that?

8. A border strip is $2\frac{1}{2}$ yards long. You cut it into pieces that are $\frac{1}{2}$ yard long. How many pieces?

Part 3: Price the Quilt

Day 3 · Use the Fabric Price Cards.

9. Find the cost of 1.25 yards of red cotton.

10. Find the cost of 2.5 yards of batting and 2 spools of thread. Add to find the total.

11. Choose two more fabrics for your quilt and decide how many yards of each to buy. Find the total cost of everything.

Final Product: Quilt Plan & Price Sheet

Your final product must include:

- A drawing of your 4 × 5 quilt with colors
- How much of each fabric you need (in yards)
- How many squares each color makes
- A price sheet with the total cost
- One sentence about your favorite part of the design

Reflection

R1. Why does dividing by $\frac{1}{8}$ give a bigger number?

R2. Where do you line up decimals when you multiply money?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Fabric amounts, cuts, and costs are computed with correct fraction and decimal operations and match the quilt design. Goes beyond the minimum with a creative, realistic choice.	Fabric amounts, cuts, and costs are computed with correct fraction and decimal operations and match the quilt design.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Quilt plan shows the diagram, fabric amounts, cutting plan, and price sheet. Polished, organized, and visually engaging.	Quilt plan shows the diagram, fabric amounts, cutting plan, and price sheet. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.